



Agenda



Introduce the scenario and the solution design

Task 0: Review architecture and infrastructure

Task 1: Implement and test the data agent

Break

Task 2: Implement and test the improvement agent

Break

Task 3: Implement and test the intervention agent

Course wrap-up

Scenario



Cymbal Meet is an enterprise videoconferencing company that sells physical conference room devices and SaaS software licenses to Enterprise, Mid-Market, and SMB customers. Like many SaaS businesses, Cymbal Meet faces a critical challenge: when customers don't fully adopt the product, they're far more likely to churn at renewal time. Customer success managers need a way to proactively identify underengaged accounts and take action before it's too late.

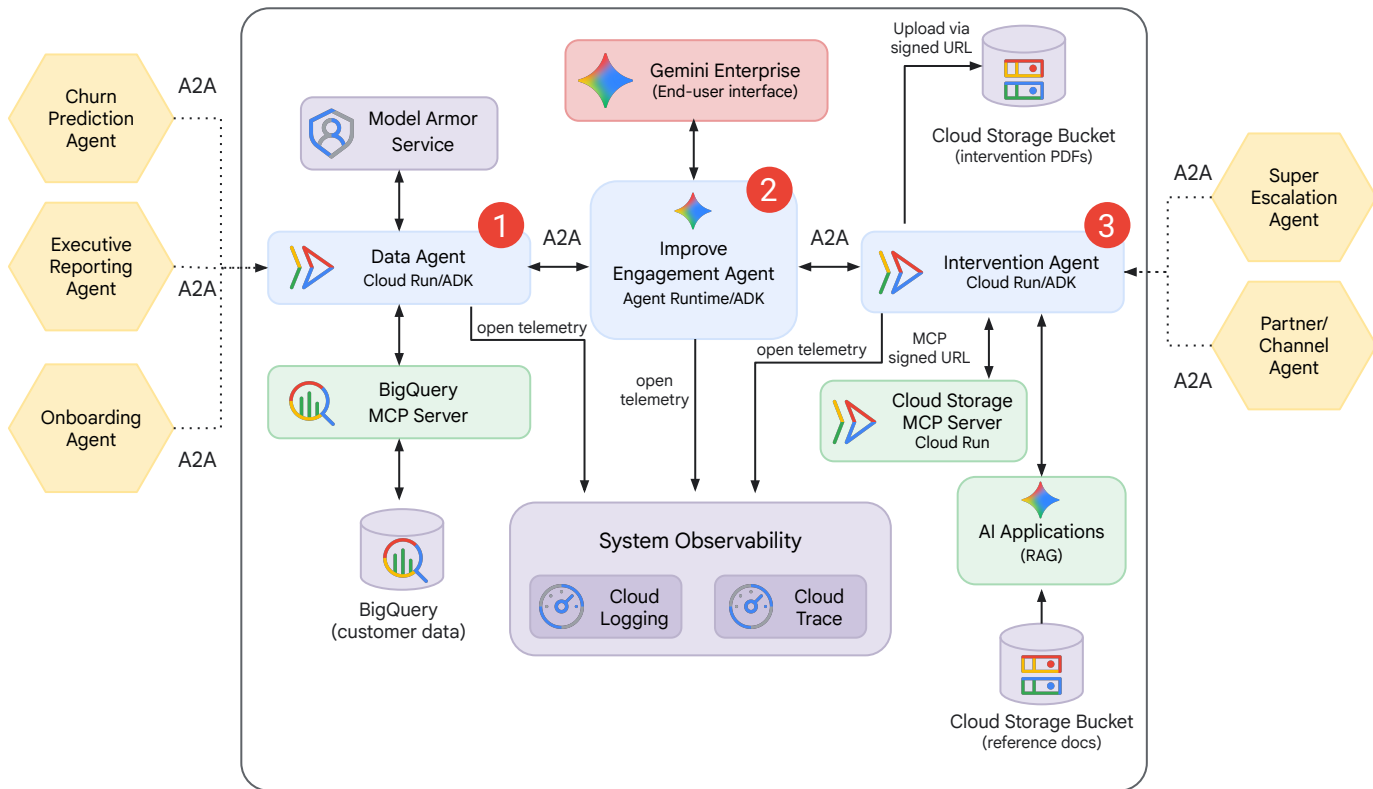
Today, identifying at-risk customers requires manually querying engagement data, cross-referencing support cases, and crafting individualized outreach—a process that's slow, inconsistent, and doesn't scale. The company needs an automated system that can analyze customer engagement metrics, surface accounts with shortfalls, and generate tailored intervention plans grounded in best practices.

Architecture



When customers underuse the product, they churn. This multi-agent system identifies at-risk accounts and generates tailored interventions -- automatically.

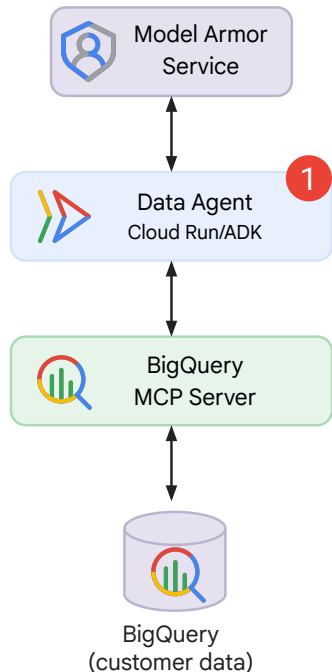
Customer Engagement Agent System



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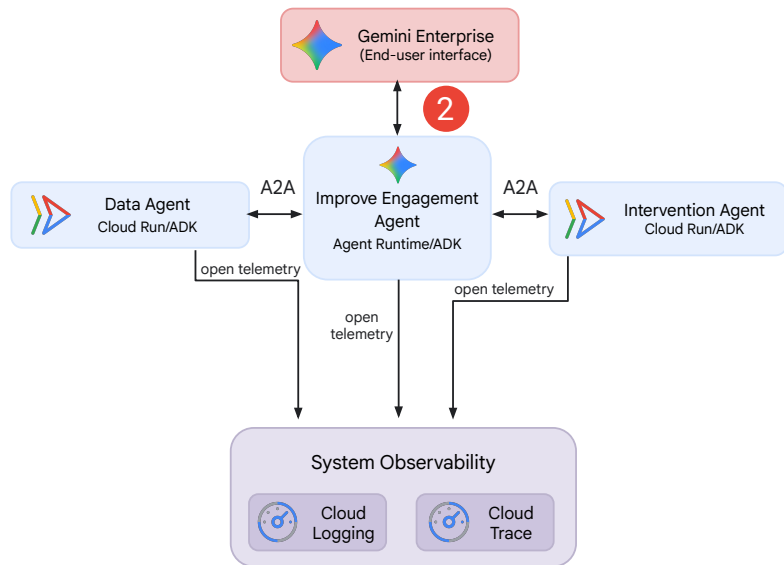
Proprietary & Confidential

Data agent



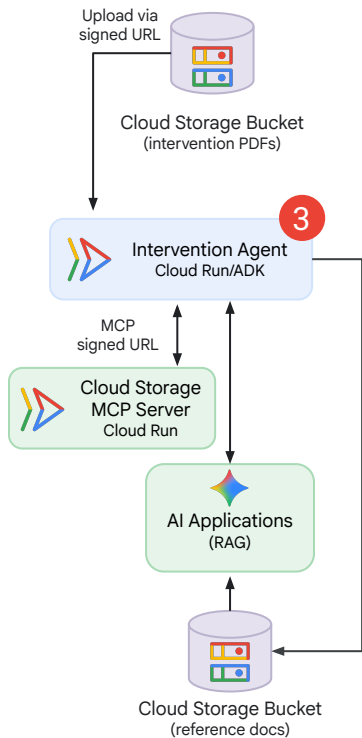
- Built with Agent Development Kit (ADK)
- Uses MCP tool to interact with BigQuery
- Understands data, composes and executes queries, returns results
- Responses are de-identified using Model Armor
- Logs and traces are written to Cloud Logging and Trace
- Wrapped in an A2A serving framework, accessed as remote agent
- Deployed to Cloud Run; secured by IAM

Improve engagement agent



- Built with Agent Development Kit (ADK)
- Orchestrates efforts by data and intervention agents
- Deployed to Agent Runtime
- Exposed to end users via Gemini Enterprise
- Logs and traces are written to Cloud Logging and Trace

Intervention agent



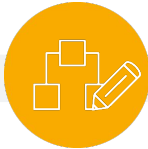
- Built with Agent Development Kit (ADK)
- Finds relevant documents using AI Applications datastore
- Reviews database data and document data, and composes customer-specific intervention plans
- Generates PDF versions of interventions
- Uses Cloud Storage MCP server to get signed URLs and uploads documents
- Logs and traces are written to Cloud Logging and Trace
- Wrapped in an A2A serving framework, accessed as remote agent
- Deployed to Cloud Run; secured by IAM

Today's tasks



**Implement and
test data agent**

Standard lab



**Implement and
test improvement
agent**

Standard lab



**Complete the
solution**

Challenge

Task 1:

Implement and test the data agent (MCP, A2A, Model Armor, OpenTelemetry, Cloud Run)

- | | |
|----------------------------|------------------------------|
| Set up dev environment | Add Model Armor sanitization |
| Review base agent code | Create an A2A server |
| Implement MCP tooling | Test locally |
| Enable logging and tracing | Deploy to Cloud Run |



15 minutes

Task 1 Discussion:

Implement and test the data agent (MCP, A2A, Model Armor, OpenTelemetry, Cloud Run)

- | | | | |
|--|----------------------------|---|------------------------------|
|  | Set up dev environment |  | Add Model Armor sanitization |
|  | Review base agent code |  | Create an A2A server |
|  | Implement MCP tooling |  | Test locally |
|  | Enable logging and tracing |  | Deploy to Cloud Run |



Share your insight!

Question 1

Name one important thing you noted about the prompt

Question 2

What were the 1-2 most important things you learned?

Question 3

What are the implications of using the agent identity to talk to BigQuery?

Question 4

How does the plugin work?

Question 5

What questions do you have?

Task 2:

Implement and test the improvement agent (OpenTelemetry, Remote Agents, Agent Tools, Agent Runtime)

- | | |
|---|--|
|  Review base agent code |  Add tools to agent |
|  Add authentication tooling |  Test locally |
|  Build RemoteA2aAgent objects |  Deploy to Agent Runtime |
|  Create AgentTools |  Add agent to Gemini Enterprise |

Task 2 Discussion:

Implement and test the improvement agent (OpenTelemetry, Remote Agents, Agent Tools, Agent Runtime)

- | | |
|---|--|
|  Review base agent code |  Add tools to agent |
|  Add authentication tooling |  Test locally |
|  Build RemoteA2aAgent objects |  Deploy to Agent Runtime |
|  Create AgentTools |  Add agent to Gemini Enterprise |



Share your insight!

Question 1

Why use the AgentTool class? What would be the alternative?

Question 2

What's happening when you do an Agent Runtime deployment?

Question 3

What are the 2-3 most interesting/important things you learned?

Question 4

What questions do you have?

Task 3:

Challenge Task: Complete, deploy, and test the intervention agent. Extend the improvement agent to use the intervention agent. Test the system end-to-end.

- | | |
|--|---|
|  Complete 2nd agent |  Create an A2A server |
|  Deploy 2nd agent |  Deploy to Cloud Run |
|  Review 3rd agent code |  Testing the entire solution |
|  Implement MCP tooling | |

Task 3:

Challenge Task: Complete, deploy, and test the intervention agent. Extend the improvement agent to use the intervention agent. Test the system end-to-end.

- | | | | |
|--|-----------------------|---|-----------------------------|
|  | Complete 2nd agent |  | Create an A2A server |
|  | Deploy 2nd agent |  | Deploy to Cloud Run |
|  | Review 3rd agent code |  | Testing the entire solution |
|  | Implement MCP tooling | | |



Share your insight!

Question 1

What are the 2-3 most important/interesting things you learned in this overall?

Question 2

What part of this do you think would really work in your company?
What won't work?

Question 3

What questions do you have?

Architecture



When customers underuse the product, they churn. This multi-agent system identifies at-risk accounts and generates tailored interventions -- automatically.

Customer Engagement Agent System

